CSE 717 — Information Security

Topic 1: Classical Ciphers

Complete Past Paper Solutions: 2020–2024

Source: *LcMaterial#1_InfoSecuritySP26.pdf* (59 pages, read in full)

Every classical-cipher question 2020–2024 included. Zero omissions.

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Quick Reference — Algorithms

The Recurring Playfair Matrix — MEMORIZE THIS

2020, 2021, AND 2023 all GIVE the exact same 5×5 matrix (I/J share a cell):

M	F	H	I/J	K
U	N	O	P	Q
Z	V	W	X	Y
E	L	A	R	G
D	S	T	B	C

High probability it recurs in 2026. If a matrix is given in the exam, you only need the encryption *procedure* below — do not waste time building a matrix from a keyword unless explicitly asked.

Caesar Cipher

Encryption: $C = (P + k) \bmod 26$ **Decryption:** $P = (C - k) \bmod 26$, where each letter is mapped A=0, B=1, …, Z=25. General affine form sometimes used: $C = (aP + b) \bmod 26$ (with a coprime to 26).

Playfair Cipher — Encryption Procedure

1. Remove spaces/punctuation, uppercase the message; treat I and J as the same letter.
2. Split into digraphs (pairs) left to right:
   - If both letters of a pair are identical, insert `X` between them and re-pair (the second of the pair shifts forward).
   - If the message has odd length, append `X` to the final single letter.
3. For each digraph (L_1, L_2) locate both letters in the matrix:
   - **Same row** → replace each letter with the letter immediately to its *right* (wrap around row).
   - **Same column** → replace each letter with the letter immediately *below* it (wrap around column).
   - **Rectangle** (different row & column) → replace each letter with the letter in its *own row* but the *other letter's column.*

Vigenère Cipher

Repeat the keyword to match the plaintext length. For position i:

$$C_i = (P_i + K_i) \bmod 26$$

(decryption: $P_i = (C_i - K_i) \bmod 26$)

One-Time Pad (OTP)

$C_i = (P_i + K_i) \bmod 26$ where K is a truly random key, **as long as the message**, used **exactly once**, and kept secret. Provides Shannon *perfect secrecy* — ciphertext reveals zero information about plaintext — but ONLY if all four conditions hold.

Hill Cipher (2×2, not yet in 2020–2024 papers but in Lc#1)

$C = (K \times P) \bmod 26$, where P is a 2×1 plaintext vector and K is an invertible 2×2 key matrix mod 26. Decryption: $P = (K^{-1} \times C) \bmod 26$, where $K^{-1} = (\det K)^{-1} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \bmod 26$ for $K = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

2024 — Q2(a,b): Caesar Encrypt + Decrypt (3 marks)

Question — 2024 Q2

(a) Encrypt the message "CRYPTO" first with a Caesar cipher shift of 5. [1.5]

(b) A message was encrypted using a Caesar cipher with the encryption function $f(p) = (p + 4) \bmod 26$. The ciphertext received is "ONXNG". Determine the original plaintext. [1.5]

Part (a): Encrypt "CRYPTO", shift $k = 5$

Caesar Encryption — 2024 Q2(a)

$C = (P + 5) \bmod 26$, letter-by-letter:

Plain	Numeric P	$P + 5$	mod26	Cipher
C	2	7	7	H
R	17	22	22	W
Y	24	29	3	D
P	15	20	20	U
T	19	24	24	Y
O	14	19	19	T

Ciphertext: `HWDUYT`

Part (b): Decrypt "ONXNG", $f(p) = (p + 4) \bmod 26$

Caesar Decryption — 2024 Q2(b)

Decryption is the inverse: $P = (C - 4) \bmod 26$.

Cipher	Numeric C	$C - 4$	mod26	Plain
O	14	10	10	K
N	13	9	9	J
X	23	19	19	T
N	13	9	9	J
G	6	2	2	C

Plaintext: `KJTJC`

Note: the question only asks you to apply the inverse function — the result need not be a dictionary word. Show the per-letter table; that is where the marks are.

2023 — Q1(a): Playfair Encryption with Given Matrix (4 marks)

Question — 2023 Q1(a)

Using the following Playfair matrix, encrypt the message: **"See you at CSECU"** [4]

M	*F*	*H*	*I/J*	*K*
U	*N*	*O*	*P*	*Q*
Z	*V*	*W*	*X*	*Y*
E	*L*	*A*	*R*	*G*
D	*S*	*T*	*B*	*C*

Playfair Encryption — 2023 Q1(a)

Step 1 — prepare the message. Remove spaces, uppercase:

`SEEYOUATCSECU` (13 letters)

Step 2 — form digraphs. Pair sequentially; for a repeated-letter pair insert `X`; pad an odd final letter with `X`:

`SE EY OU AT CS EC UX`

Step 3 — apply the row/column/rectangle rule to each digraph:

Digraph	Rule applied	Cipher
SE	Rectangle: S(row4,col1)→col0=D; E(row3,col0)→col1=L	DL
EY	Rectangle: E(row3,col0)→col4=G; Y(row2,col4)→col0=Z	GZ
OU	Same row (U N O P Q): O→right=P; U→right=N	PN
AT	Same column (H O W A T): A→below=T; T→below(wrap)=H	TH
CS	Same row (D S T B C): C→right(wrap)=D; S→right=T	DT
EC	Rectangle: E(row3,col0)→col4=G; C(row4,col4)→col0=D	GD
UX	Rectangle: U(row1,col0)→col3=P; X(row2,col3)→col0=Z	PZ

Ciphertext: `DL GZ PN TH DT GD PZ` → `DLGZPNTHDTGDPZ`

Common mistakes

- Forgetting to recombine "See you at" → `SEEYOUAT` *without spaces* before pairing — spaces are NOT encrypted as letters.
- Mixing up the rectangle rule direction: each letter takes *its own row, the other letter's column* — not the reverse.
- Forgetting the wrap-around on same-row/same-column rules (e.g. rightmost column wraps to leftmost, bottom row wraps to top).

2022 — Q2(b): Caesar / OTP Cryptanalysis (conceptual, part of 10-mark Q2)

Question — 2022 Q2

(a) What are the essential ingredients of a symmetric cipher? [3]

(b) How can cryptanalysis possibly work for Caesar cipher and one-time pad? [3]

(c) What are the principal elements of a public-key cryptosystem? What are the roles of the public and the private key? [4]

Part (b): Cryptanalysis of Caesar Cipher and OTP

Conceptual Answer — 2022 Q2(b)

Caesar cipher — trivially broken:

- **Brute force / exhaustive key search:** only 25 possible keys ($k = 1..25$) — try all shifts, the one producing readable text is correct. Computationally negligible.
- **Frequency analysis:** the shift preserves the relative frequency distribution of letters (e.g. `E` remains the most frequent letter, just relabelled). Comparing ciphertext letter frequencies to known English letter frequencies reveals the shift k directly.

One-time pad — cryptanalysis only works if misused:

- If used *correctly* (truly random key, as long as the message, used once, kept secret) OTP gives **perfect secrecy** — the ciphertext is statistically independent of the plaintext, so *no* amount of computation or ciphertext can reveal any information. Cryptanalysis is mathematically impossible.
- Cryptanalysis "works" only when one of the conditions is violated:
  - **Key reuse** → XOR-ing two ciphertexts cancels the key, leaking $P_1 \oplus P_2$ (crib-dragging attacks).
  - **Non-random key** (e.g. generated by a weak PRNG or derived from text) → key itself becomes analyzable.
  - **Key shorter than message** (i.e. effectively a Vigenère) → periodicity allows frequency analysis per key-position.

One-line answer: Caesar falls to brute force/frequency analysis because the keyspace and structure are tiny; OTP cryptanalysis only succeeds when the "one-time", "random", or "secret" guarantees are broken.

2021 — A2: OTP Discussion + Playfair (4.75 marks); A4(iv): Caesar Program (2 marks)

Question — 2021 A2 & A4(iv)

A-2.

(a) (i) The one-time pad offers complete security — do you agree on this? Defend your answer. (ii) What are the fundamental difficulties of the one-time pad? [2.75]

(b) Using the given Playfair matrix (same as above), encrypt the message: "See you at CSE department". [≈2]

A-4(iv). Write a program that can encrypt and decrypt using the general Caesar cipher. [2]

Part A2(a): OTP — "Complete Security" Discussion

OTP Conceptual Defence — 2021 A2(a)

(i) Do you agree? Agree, but only under strict conditions. Shannon proved OTP achieves *perfect secrecy*: $H(P \mid C) = H(P)$ — the ciphertext gives an attacker zero information about the plaintext, *provided*:

1. the key is **truly random** (not pseudo-random),
2. the key is **at least as long as** the message,
3. the key is used **exactly once** and never reused, and
4. the key is kept **completely secret** between sender and receiver.

If even one condition fails, the security guarantee collapses entirely.
(ii) Fundamental difficulties (why OTP is impractical):

- **Key generation:** producing truly random bits at the required volume is hard.
- **Key distribution:** a key as long as the message must be shared *in advance, securely* — as hard as securing the message itself.
- **Key storage:** both parties must store, protect, and synchronize huge key material.
- **No reuse:** every new message needs fresh key material — keys cannot be cached or compressed.
- **Scalability:** impractical for high-volume or long-running communication (e.g. internet traffic).

Part A2(b): Playfair — "See you at CSE department"

Playfair Encryption — 2021 A2(b)

Step 1. Remove spaces, uppercase: `SEEYOUATCSEDEPARTMENT` (21 letters).
Step 2. Digraphs (no repeated-letter pairs occur; final letter `T` is alone → pad with `X`):

`SE EY OU AT CS ED EP AR TM EN TX`

Step 3. Apply rules (first five digraphs reuse the 2023 Q1(a) results):

Digraph	Rule applied	Cipher
SE	(as 2023 Q1a) Rectangle	DL
EY	(as 2023 Q1a) Rectangle	GZ
OU	(as 2023 Q1a) Same row	PN
AT	(as 2023 Q1a) Same column	TH
CS	(as 2023 Q1a) Same row	DT
ED	Same column (M U Z E D): E→below=D; D→below(wrap)=M	DM
EP	Rectangle: E(row3,col0)→col3=R; P(row1,col3)→col0=U	RU
AR	Same row (E L A R G): A→right=R; R→right=G	RG
TM	Rectangle: T(row4,col2)→col0=D; M(row0,col0)→col2=H	DH
EN	Rectangle: E(row3,col0)→col1=L; N(row1,col1)→col0=U	LU
TX	Rectangle: T(row4,col2)→col3=B; X(row2,col3)→col2=W	BW

Ciphertext: `DL GZ PN TH DT DM RU RG DH LU BW` → `DLGZPNTHDTDMRURGDHLUBW`

Part A4(iv): Caesar Cipher Program

Python Program — General Caesar Encrypt/Decrypt — 2021 A4(iv)

```
def caesar_encrypt(text, k):
    result = ""
    for ch in text.upper():
        if ch.isalpha():
            result += chr((ord(ch) - ord('A') + k) % 26 + ord('A'))
        else:
            result += ch
    return result

def caesar_decrypt(text, k):
    return caesar_encrypt(text, -k)

# Example
print(caesar_encrypt("CRYPTO", 5))   # -> HWDUYT
```

```
print(caesar_decrypt("HWDUYT", 5))   # -> CRYPTO
```

Exam tip: if asked to "write a program", pseudocode or Python both earn marks — the examiner is checking you understand the modular-arithmetic loop, not syntax perfection.

2020 — Q1(a,b): Playfair Encryption (4 marks) + Vigenère Encryption (4.75 marks)

Question — 2020 Q1

(a) Using the given Playfair matrix (same matrix as above), encrypt this message: "Must see you over CU playground. Coming now." [4]

(b) Using the Vigenère cipher, encrypt the word "explanation" using the key "cse". [4.75]

Part (a): Playfair — "Must see you over CU playground. Coming now."

Playfair Encryption — 2020 Q1(a)

Step 1. Strip punctuation/spaces, uppercase:

`MUSTSEEYOUOVERCUPLAYGROUNDCOMINGNOW` (35 letters)

Step 2. No adjacent-pair repeats occur; pair sequentially and pad the trailing single letter `W` with `X`:

`MU ST SE EY OU OV ER CU PL AY GR OU ND CO MI NG NO WX` (18 digraphs)

Step 3. Apply the row/column/rectangle rule:

Digraph	Rule applied	Cipher
MU	Same col (M U Z E D): M→below=U; U→below=Z	UZ
ST	Same row (D S T B C): S→right=T; T→right=B	TB
SE	Rectangle (as before)	DL
EY	Rectangle (as before)	GZ
OU	Same row (as before)	PN
OV	Rectangle: O(row1,col2)→col1=N; V(row2,col1)→col2=W	NW
ER	Same row (E L A R G): E→right=L; R→right=G	LG
CU	Rectangle: C(row4,col4)→col0=D; U(row1,col0)→col4=Q	DQ
PL	Rectangle: P(row1,col3)→col1=N; L(row3,col1)→col3=R	NR
AY	Rectangle: A(row3,col2)→col4=G; Y(row2,col4)→col2=W	GW
GR	Same row (E L A R G): G→right(wrap)=E; R→right=G	EG
OU	Same row (as before)	PN
ND	Rectangle: N(row1,col1)→col0=U; D(row4,col0)→col1=S	US
CO	Rectangle: C(row4,col4)→col2=T; O(row1,col2)→col4=Q	TQ
MI	Same row (M F H I/J K): M→right=F; I/J→right=K	FK
NG	Rectangle: N(row1,col1)→col4=Q; G(row3,col4)→col1=L	QL
NO	Same row (U N O P Q): N→right=O; O→right=P	OP
WX	Same row (Z V W X Y): W→right=X; X→right=Y	XY

Ciphertext: `UZTBDLGZPNNWLGDQNRGWEGPNUSTQFKQLOPXY`
Exam tip: with 35 letters this is the longest Playfair question seen 2020–2024 (4 marks). Work systematically left-to-right in a table like above — partial credit is given per-digraph, so do not skip steps even under time pressure.

Part (b): Vigenère — "explanation", key "cse"

Vigenère Encryption — 2020 Q1(b)

Plaintext: `EXPLANATION` (11 letters). Repeat key `CSE` to length 11: `CSECSECSECS`. $C_i = (P_i + K_i) \bmod 26$:

Plain	P_i	Key	K_i	$P_i + K_i$	mod26	Cipher
E	4	C	2	6	6	G
X	23	S	18	41	15	P
P	15	E	4	19	19	T
L	11	C	2	13	13	N
A	0	S	18	18	18	S
N	13	E	4	17	17	R
A	0	C	2	2	2	C
T	19	S	18	37	11	L
I	8	E	4	12	12	M
O	14	C	2	16	16	Q
N	13	S	18	31	5	F

Ciphertext: `GPTNSRCLMQF`

Exam Strategy Summary

High-Yield Checklist for Classical Ciphers (5/5 years, 2–9 marks)

1. **Caesar encrypt + decrypt** with arbitrary $f(p) = (ap + b) \bmod 26$ — both directions, every year since 2024 and 2021's program question. Practice the per-letter table format.
2. **Playfair with a GIVEN matrix** — memorize the recurring matrix (2020/2021/2023). Drill: prepare message (strip spaces/punctuation, handle repeats with X, pad odd length), then row/column/rectangle rules. Practice on sentences *with spaces* since that's the exam style.
3. **Vigenère** (2020) — short keyword repeated, $C_i = (P_i + K_i) \bmod 26$. Mechanical, fast marks.
4. **OTP conceptual** (2021, 2022) — be ready to argue "perfect secrecy under 4 strict conditions" and list the practical difficulties / cryptanalysis-only-on-misuse points.
5. **Hill Cipher** (Lc#1 only, no past-paper appearance) — light pass: know $C = (KP) \bmod 26$ and be able to reproduce the worked "HI"→"TC" example if time allows, but do not over-invest.